

0/24 Questions Answered

hw9-regression-mle-2003movies

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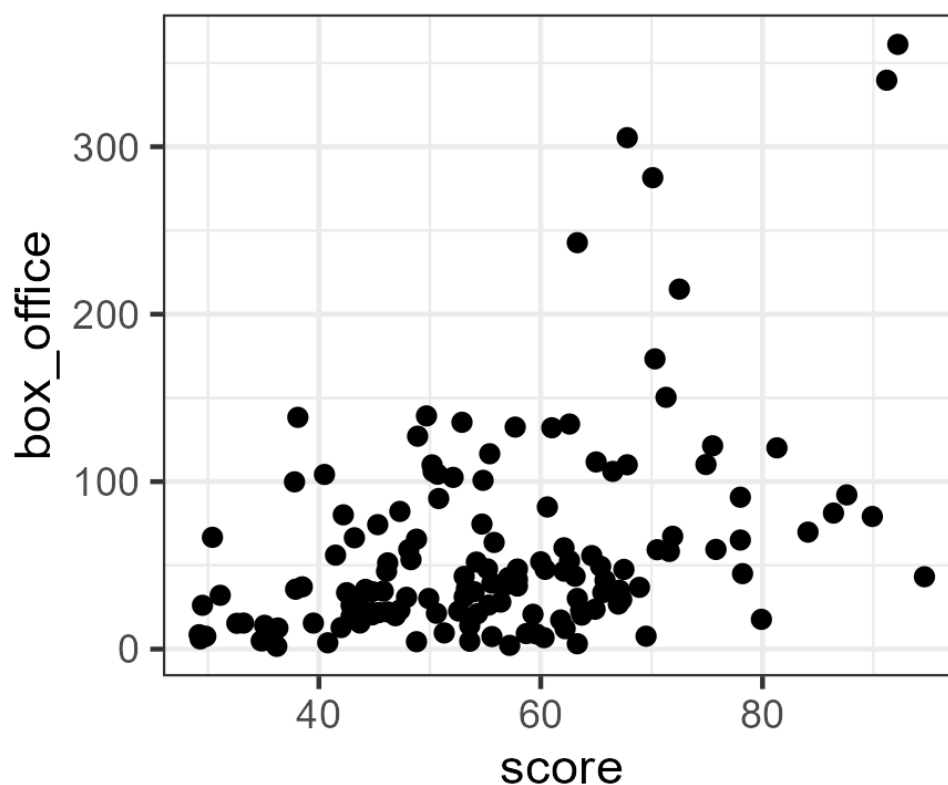


Q1

10 Points

Note: There are three options for homework 9, that differ by data topic, and you only need to complete one of the three assignments.

Do critics opinions match public opinion when it comes to movies? We have data for movies released in 2003 and we want to see if the amount that a movie earns in theaters, box office earnings (millions \$), is related to critics' opinions of the movie.



First, install the R packages if you haven't.

```
install.packages(c("broom", "modelr", "openintro"))
```

Then run the following R code to load the data set:

```
library(tidyverse)
```

```
movies_data <- openintro::movies
```

Q1.1**1 Point**

Fit a linear model to regress `box_office` on `score`, saving your model as `movie_model`. Use `broom::tidy(movie_model)` to summarize your model.

Upload a picture of both your code to fit the model and the model results.

 Please select file(s)

Select file(s)

Save Answer

Q1.2**1 Point**

What is the estimate of the intercept? Round to two decimal places.

Save Answer

Q1.3**1 Point**

What is the estimate of the slope? Round to two decimal places.

Save Answer

Q1.4**1 Point**

We want the 95% confidence interval for the **slope**.

Provide the confidence interval lower bound rounded to two decimal places.

Provide the confidence interval upper bound rounded to two decimal places.

[Save Answer](#)**Q1.5****1 Point**

Provide an interpretation of the 95% confidence interval for the slope in context.

[Save Answer](#)

Q1.6**1 Point**

What is the predicted box office earnings for a movie that received a critics score of 60.6? Round your answer to two decimal places.

[Save Answer](#)**Q1.7****1 Point**

The movie Brother Bear had a critics score of 60.6 and earned 84.874 million dollars at the box office. What is the residual for this data point, rounded to two decimal places?

[Save Answer](#)**Q1.8****1 Point**

Use `broom::glance(movie_model)` on your model and upload a picture of the output.

 Please select file(s)[Select file\(s\)](#)[Save Answer](#)

Q1.9**1 Point**

Identify the R^2 value for your model and provide an interpretation in context.

[Save Answer](#)**Q1.10****1 Point**

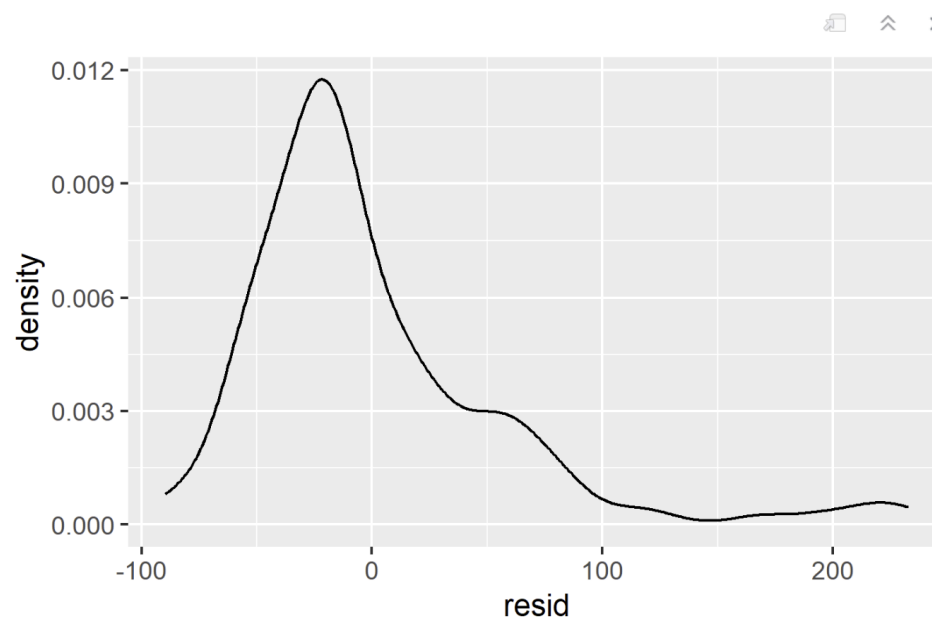
Use the `modelr::rmse(movie_model, data = movies_data)` to compute the root mean squared error. Round your answer to two decimal places.

[Save Answer](#)

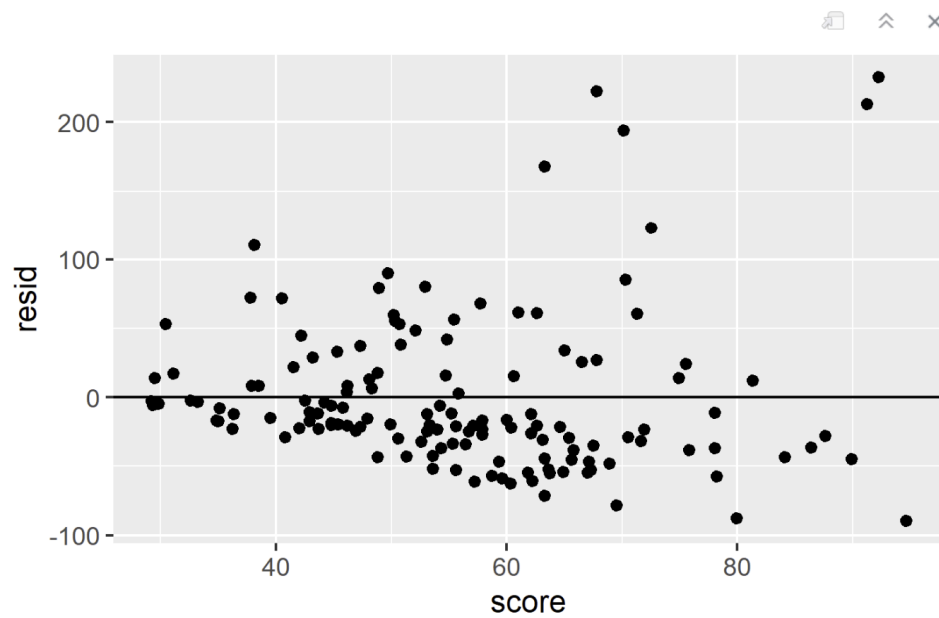
Q2**4 Points**

Now we are going to check the model conditions to make sure that our regression model was appropriate for studying the linear relationship between our response and explanatory variables. Below are a few plots to help with checking these conditions.

```
{r}  
movies_data <- movies_data |>  
  modelr::add_residuals(movie_model)  
  
ggplot(movies_data, aes(x = resid)) +  
  geom_density()
```




```
{r}  
ggplot(movies_data, aes(x = score, y = resid)) +  
  geom_point() +  
  geom_hline(yintercept = 0)
```



Q2.1

1 Point

Is the linearity condition met? Explain.

Save Answer

Q2.2**1 Point**

Is the normality condition met? Explain.

[Save Answer](#)**Q2.3****1 Point**

Is the constant variance assumption met? Explain.

[Save Answer](#)**Q2.4****1 Point**

Is the independence assumption met? Explain.

[Save Answer](#)

Q3**2 Points**

Maybe the relationship between box office earnings and critics score differs by movie rating (i.e. G, PG, PG-13, or R). We are going to add another predictor, `rating`, to our model.

Q3.1**1 Point**

Fit a linear model to regress `box_office` on `score` and `rating`, saving your new model as `movie_model2`. Use

`broom::tidy(movie_model2)` to summarize your model.

Upload a picture of both your code to fit the model and the model results.

 Please select file(s)**Select file(s)****Save Answer****Q3.2****1 Point**

Which model explains the variation in box office earnings better? Explain your answer with an appropriate model evaluation metric.

Save Answer

Q4**2 Points**

Let's check our understanding about a few other regression topics discussed in class.

Q4.1**1 Point**

When model conditions are not met, there are some strategies that might help, such as transforming your response variable. Select the scenario where transforming your response variable would definitely **not** be helpful.

When there is a nonlinear relationship between your response and predictor(s)

When your residuals are not normally distributed

When there is a pattern in the residuals, such as non constant variance

When your data are correlated

Save Answer

Q4.2**1 Point**

What strategy can help you avoid overfitting your data?

Compare the R^2 value for each model you are considering

Compare the root mean squared error value for each model you are considering

Split the data into training and testing data

Fit a model with all of the variables and keep the predictors with the lowest p-values

Save Answer

Q5**3 Points**

We are now moving onto questions pertaining to maximum likelihood estimation. For each of the following scenarios, specify if we are investigating the probability or likelihood.

Q5.1**1 Point**

The percentage of vegetarians in a random sample of 100 UCI students was 18%. What is the chance that 25% or more are vegetarians among all anteaters?

probability

likelihood

Save Answer

Q5.2**1 Point**

If the number of lightnings during one hour of storm follows a Poisson Distribution with $\lambda = 5$ what is the chance that 4 lightnings fall in 30 minutes?

probability

likelihood

Save Answer

Q5.3**1 Point**

If the sample average is 50, what is the best estimate that we can make for μ in a Normal Distribution with standard deviation 1?

probability

likelihood

Save Answer

Q6**5 Points**

Let X be a random variable with the pdf

$$f(x) = \theta x^{\theta-1} \text{ where } 0 < x < 1 \text{ and } \theta > 0$$

Let $\{0.1, 0.2, 0.3\}$ be the three observed values of X . Find the maximum likelihood estimate of θ . Round to two decimal places.

Upload a picture of your work:

 Please select file(s)

Select file(s)

Save Answer

Q7**1 Point**

The following two questions are related to the homework choice you made in choosing to do this assignment instead of the other two options. There are no correct answers, this question is only being graded for completeness (note that answering "I felt like it" or "it was interesting" is an insufficient response).

Please explain in detail what led you to choose this topic, 2003 movies, to investigate.

What was it about the other two options (gorillas and bikeshares) that led you not to choose them? Please explain in detail.

Save Answer

Q8**1 Point**

- Have you done the homework by yourself, with a peer(s), or used tools like ChatGPT? If you do not acknowledge any help from a person or ChatGPT, we will consider this cheating.

- If you provided help to a peer(s) and/or got help from your peers(s) please explain briefly what this entailed.

- If you used ChatGPT, include the prompts you used and explain how ChatGPT supported your learning.

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